EIS analysis report

Model: Randles (Rs + Rct‖Cdl)

Data file: randles_Rs10_Rct30_C100mF.csv

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1. Equivalent-circuit fit

Electrical parameters of the chosen model, found by least-squares fitting. They describe the circuit, not yet the chemistry.

R_s: 10 Ω

R_{ct}: 29.98 Ω

C_{dl}: 0.0001002 F

RMSE (complex): 0.06258 Ω

χ² (σ = 1): 3.9

2. Measured EIS features (no model)

Read directly from the data. If the system behaves like a Randles circuit: $R_{high} \approx R_s$, $R_{low} - R_{high} \approx R_{ct}$, and f_{apex} is related to R_{ct} and C_{dl} as below.

R_{high}: 10.11 Ω

R_{low}: 39.89 Ω

$R_{low} - R_{high}$: 29.78 Ω

f_{apex}: 52.11 Hz

$$f_{apex} = \frac{1}{2\pi \, R_{ct} \, C_{dl}}$$

LaTeX: `f_{apex} = \frac{1}{2 \pi \, R_{ct} \, C_{dl}}`

3. Electrochemical interpretation

Derived quantities. Each one is valid only under the assumptions listed with it.

Inputs provided

- Temperature: 25 °C
- Electrode area: 1 cm²
- Number of electrons n: 1
- Electrode spacing L: 1 cm
- Electrolyte cross-section: 1 cm²

Shared assumptions

- A1: The fitted circuit is $R_s + (R_{ct} \parallel C_{dl})$: one semicircle, and C_{dl} is an ideal capacitor (no CPE / non-ideal behaviour).

Characteristic frequency f_c

Value: 52.98 Hz

$$f_c = \frac{1}{2\pi\tau}$$

LaTeX: `f_c = \frac{1}{2 \pi \tau}`

Relies on shared assumption(s): A1

Notes:

- Should match the apex frequency measured directly from the Nyquist plot (f_{apex}); a large mismatch means the single-semicircle model does not describe the data well.

Specific double-layer capacitance

Value: 0.0001002 F/cm^2

$$C_{spec} = \frac{C_{dl}}{A}$$

LaTeX: `C_{spec} = \frac{C_{dl}}{A}`

Relies on shared assumption(s): A1

Additional assumptions:

- A is the true electroactive area, not just the geometric area.
- The fitted capacitance is really the double layer (not adsorption pseudocapacitance or a coating).

Notes:

- 100 µ F/cm^2 is outside the usual ~5-100 µ F/cm^2 range for flat electrodes in aqueous electrolyte. Possible reasons: real area differs from the area you entered (roughness, porosity), a surface film or functionalised layer, a non-aqueous electrolyte, or a non-ideal (CPE) interface. EIS alone cannot tell which.

Exchange current density j_0

Value: 0.0008569 A/cm^2

$$j_0 = \frac{RT}{nFR_{ct}A}$$

LaTeX: `j_0 = \frac{R\,T}{n\,F\,R_{ct}\,A}`

Relies on shared assumption(s): A1

Additional assumptions:

- Measured at equilibrium (open-circuit) with both the oxidised and reduced species present, small AC amplitude (linearised Butler-Volmer).
- R_{ct} is purely the charge-transfer resistance: no diffusion (Warburg) or film resistance mixed into it.
- Single, simple electron-transfer reaction; n is the number of electrons of that reaction (ambiguous for multi-step mechanisms).
- A is the true electroactive area.
- T = 298.15 K.

Notes:

- If R_{ct} also contains other resistive contributions, j_0 is underestimated.

Consistency check: fit vs data

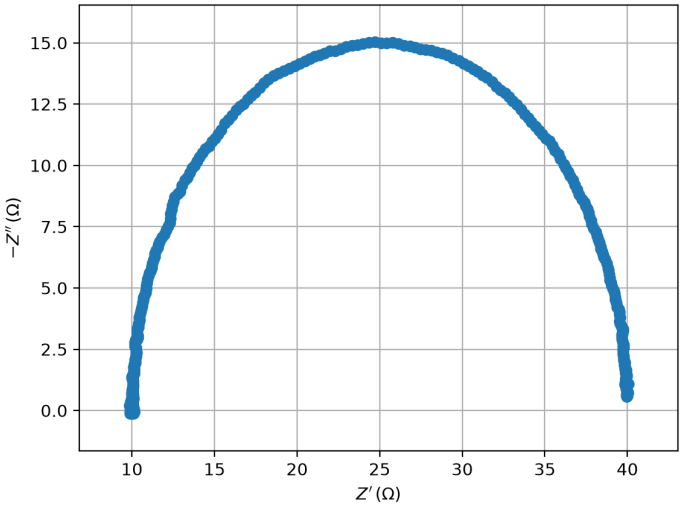
f_c (fit): 52.98 Hz

f_{apex} (data): 52.11 Hz

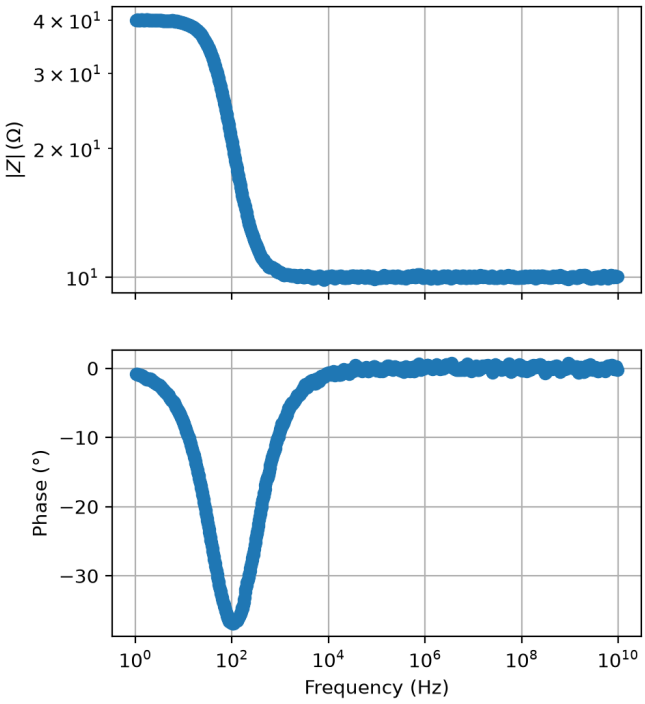
Difference: 1.7 %

The fit and the data agree on the time constant. This supports the single-semicircle model but does not prove it.

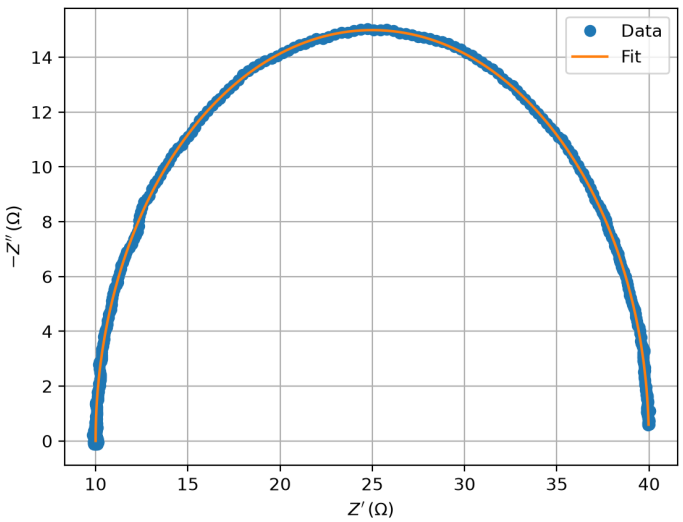
4. Figures



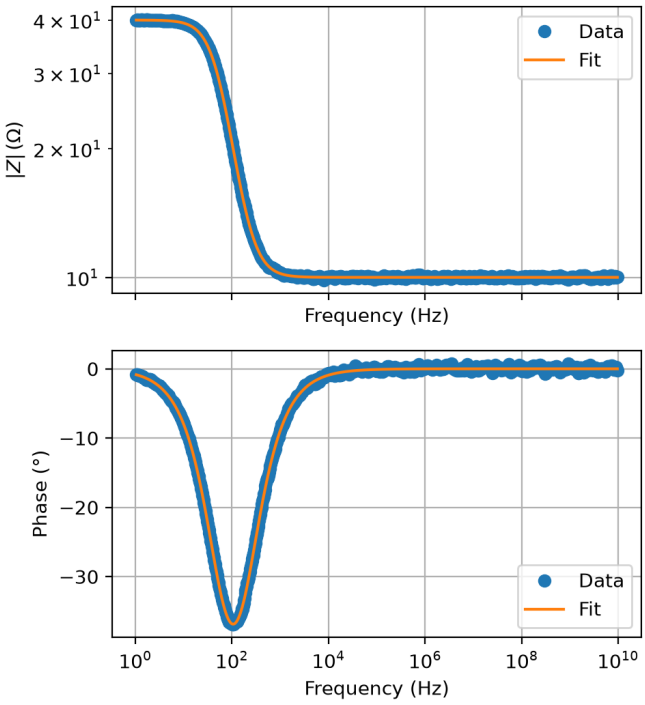
Nyquist plot (data)



Bode plot (data)



Nyquist: data vs fitted model



Bode: data vs fitted model